

MENU 3 - SCALE

(POS. 3.1-2) Flow Rate Full Scale 1-2 [FS1-2= l/s xxxx.x] AL2 [FRFS1-FRFS2]

The FS2 (full scale flow rate 2) is active with POS. 6.6 see page 28 ,enabled.

The full scale is used to indicate to the maximum meter's flow rate. The full scale should be chosen carefully as it's parameters are used for several other parameters. There are three fields to fill in order to set this parameter, from left to right: 1) measure unit, 2) time unit of measure and 3) numeric value. The selection is made by positioning the cursor on the field to modify. To change the type unit of measure (metric, Imperial units, mass or volume) see POS. 2.3 see page 27 and POS. 2.19 see page 27 and POS. 2.20 see page 27. The value of Fs1-2 depend nominal diameter POS. 1.4 see page 27. The following tables shown the units of measure available and the conversion factor by comparison with 1dm³ and 1kg. The converter accepts any kind of combination of units of measure satisfying both the following conditions:

□ Numeric field value 99999

□ $1/25 \text{ fsmax} \leq \text{numeric field value} \leq \text{fsmax}$.

Where fsmax is the maximum full scale value corresponding to the sensor, equal to a 10m/s liquid speed. The measure units are shown as appear on the display. The Imperial units units are diversified by using capital and small characters.

METRIC	
cm³	Cubic centimeter
ml	Milliliter
l	Liter
dm³	Cubic decimeter
dal	Decalitre
hl	Hectolitre
m³	Cubic meter
ML	Mega Liter

NOT METRIC	
in³	Cubic inch
Gal	American gallon
ft³	Cubic foot
bbl	Standard barrel
BBL	Oil barrel
hf³	Hecto cubic feet
KGL	American Kilo gallon
kf³	Kilo cubic feet
ttG	Ten thousand gallons
IGL	Imperial gallon
IKG	Imperial Kilo gallon
Aft	Acre foot
MGL	Mega gallon
IMG	Imperial mega gallon

MASS UNIT NOT METRIC	
Oz	Ounce
Lb	Pound
Ton	Short tons

MASS UNIT METRIC	
g	Gram
kg	Kilogram
t	Ton

When a measure mass unit is set, the specific gravity function is automatically enabled by the system. Please, note that the mass measure is heavily affected by the temperature. With certain liquids this may cause significant measurement errors. The following measure of time units can be selected: **s** = second, **m** = minute, **h** = hour, **d** = day .

NOTES FOR USING THE MCP INTERFACE

The command FRFS1 =? and command FRS2 = ?, edited by MCP software, return a list of only the unit compatible with the nominal diameter set. If the sensor is insertion type and the diameter is zero, the only possible unit is m/s if the flow rate were chosen metric units, else f/s for the unit of measurement non metric.

(POS. 3.2) *Batching quantity***AL2****[BTQTY]**

This function is enabled when output1 is set to "BATCHING".

It allows to set the unit of measure and the batching quantity.

(POS. 3.3-3.5) *Output Pulse 1-2* [Pls1-2= dm³ x.xxxxx]**AL2****[OP1PV-OP2PV]**

Pls1 and Pls2 is active with POS. 7.1 see page 29 and POS. 7.2 see page 29, enable and setting pulse value on channel 1 and channel 2. This function allows the user to set a signal (a pulse) to be given from the converter when a defined amount of liquid has passed through the sensor. To set the parameter, complete the 2 fields, from left to right: 1) measure unit, 2) numeric value. The selection is performed by positioning the cursor in the field to be modified. To change the unit type (metric, British or American, mass or volume) see POS. 2.4 see page 27 and POS. 2.5 see page 27, POS. 2.19 see page 27 and POS. 2.20 see page 27. The value of Pls1-2 depends on nominal diameter POS. 1.4 see page 27. Only those units described (POS. 3.1 see page 28-POS. 3.3 see page 28) above are available to be selected.

(POS. 3.4-3.6) *Output1-2 Pulse Time* [Tpls1-2= ms x.xxxxx]**AL2****[OP1PT OP2PT]**

Tpls1 and Tpls2 is active with POS. 7.1 see page 29 and POS. 7.2 see page 29 enable. Setting duration of the pulse generated on channel 1 and 2.

With the liquid volume to generate the pulse value (POS. 3.4 see page 28 POS. 3.6 see page 28) set by the user. The user must set the corresponding duration of the pulse to be outputted. This value is expressed in milliseconds and has to be between 0.4 and 9999.99. When the high frequency output is present, then the minimum value can type of device is connected to the converter, the user must verify that the set pulse duration is compatible with the external device processing such pulses. If, for example, an electro-mechanical pulse counter is connected, a minimum pulse time of 0.04 milliseconds can be set.

ATTENTION: The converter can not detect problems that may occur; firstly, the pulse is too long the coils may burn out, secondly, if the pulse is too short, the counter may not be able to function, causing damage of the output.

(POS. 3.7-3.8) *Output full scale freq.1-2***[Frq1-2= x.xxxxx]****AL2****[OU1FF-OU2FF]**

Frq1 and Frq2 are activated with POS. 7.1 see page 29 and POS. 7.2 see page 29 enabled and set to the value freq+/-/+-. Setting duration of the pulse generated on channel 1 and 2.

MENU 4 - MEASURE

(POS. 4.1) Damping

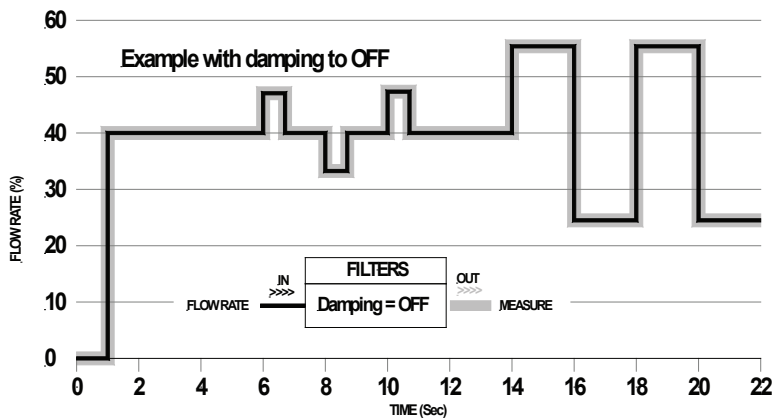
[Damping=OFF/SMART/(TIME)]

AL3

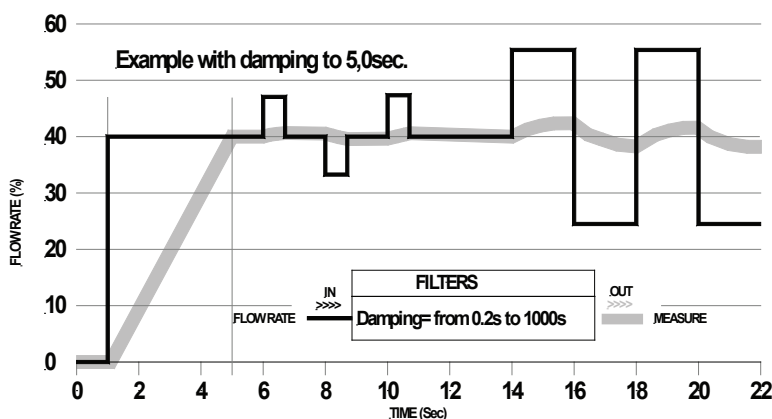
[MFDMP]

This section of manual is extremely important because the correct setting of the filters allows to obtain a proper response of the instrument to the measured flow rate and the specific requirements of use; as a general rule, consider that, starting from Damping = OFF (no filter applied to the measure), successive values, introduce increasing damping. The following diagrams show the instrument's response to changes in flow rate from 0 to 100%, using the different settings of the damping function.

The SMART is an adaptive filter that adapts automatically to most of the processes (recommended value), making the response of the meter very ready to fast changes of flow and at the same time extremely precise and stable for slow variations. It may be convenient to use a constant damping filter time, where there is a pulsating flow (for example generated by peristaltic pumps). With longer times you get a mean value stable, while with short times the measure will closely follow the flow pulses, but consequently more unstable. NOTE: If the rechargeable battery is active, the damping could be set only in "SMART".



Damping function(OFF). the meter follows the trend of fast changes in flow.



Damping mode based on time (fro 0.2s to 1000s) The measure is averaged over a number of samples determined by the value assigned to the dampening function. When the damping parameter is expressed in seconds, the filter works damping the measurement noise and sudden change of flow rate. Increasing the parameter of damping increases the stability of the measurement.

(POS. 4.2) Cut-off threshold**[Cut-off=% xxx]****AL3****[MFCUT]**

Setting the low flow cutoff threshold. This function is useful to avoid that flows close to zero, due to the electrical noises from tiny movements of liquid (due for example to vibrations of the pipe) which cause an increasing of the totalizers. The allowed range for this function is 0-25% of full scale set. For most applications a value between 0.5 and 1% is recommended.

(POS. 4.3) Calibration verify**[Cal.verify=ON]****AL3****[ACAVE]**

This function enables an automatic verification of board's coefficients. As the converter performs continuously a large number of tests, we recommend to use this function only in presence of wide range of temperature. Instead it is NOT recommended to use it when the instrument is used in metering applications (batch).

(POS. 4.4) Automatic scale change enable**[Autorange=ON/OFF]****AL3****[ARNGE]**

Enables the automatic change of scale. The meter may have two different working ranges in order to suit to the variable process conditions. In order to get the best results out of this function it is important range N.2 (Fs2) if enabled is bigger than N.1 (Fs1). When the flow rate increases and reaches the 100% of the full scale 1, then the meter automatically switches to scale 2. When the flow rate decreases again reaching a value on scale 2 equal to the 90% of full scale N.1, then the active scale is 1 again. Allowed values for this parameter: ON/OFF.

Notice: when the autorange is enabled, is not allow to use the manual range change (POS. 6.6 see page 28).

This function does NOT increase the accuracy of the measure; its aim is to increase the resolution of 4/20 mA when the meter work at very low flow rates (typical case the flow rate of water distribution with daytime flow much higher than the night flow). N.B: this function is disabled if the batching function is active

(POS. 4.5) High immunity INPUTS**[H.imm. inp.]****AL4****[HIINP]**

The HIINP function (INPut High Immunity filter) introduces a hardware filter to be used ONLY IN CASE OF ABSOLUTE NECESSITY, when the measure is absolutely unstable or it is NOT possible to make the measure, and every possible attempt to reduce or eliminate the noise do not give a positive result, with particular attention of instrument ground connection. When this function is activated (HIINP = ON) the measure will be influenced by an unavoidable error estimated around 1%.

MENU 4 - MEASURE: ONLY MCP FUNCTIONS**Measure Filter Cut-off Threshold 2****[MCP ONLY]****AL3****[MFCT2]**

Setting the low flow cutoff threshold, it is similar to the function in 4.2. The value of this function is NOT visible on display but only with MCP command.

Dynamic sample analysis**[MCP ONLY]****AL4****[DYNSA]**

Reserved to the service

Dynamic sample time**[MCP ONLY]****AL4****[DYNST]**

Reserved to the service

MENU 5 - ALARMS

(POS. 5.1) *Maximum direct flow rate threshold* **[Max+=XXXXX]** **AL3 [FRAXP]**

Maximum value alarm set for direct flow rate set (expressed in technical units). When the flow rate value exceeds such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.2) *Maximum reverse flow rate threshold* **[Max-=XXXXX]** **AL3 [FRAXN]**

Maximum value alarm set for reverse flow rate set (expressed in technical units). When the flow rate value exceeds such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.3) *Minimum direct flow rate threshold* **[Min+=XXXXX]** **AL3 [FRANP]**

Minimum value alarm set for reverse flow rate set (expressed in technical units). When the flow rate value falls below such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.4) *Minimum reverse flow rate threshold* **[Min-=XXXXX]** **AL3 [FRANN]**

Minimum value alarm set for reverse flow rate set (expressed in technical units). When the flow rate value falls below such a threshold, then an alarm message is generated. Setting this parameter to zero disables the alarm generation.

(POS. 5.5) *Hysteresis* **[Hysteresis=% XX]** **AL3 [ATHYS]**

Hysteresis threshold set for the minimum and maximum flow rate alarms. The value of this parameter is expressed as percentage of the full scale value and may be set from 0 to 25%.

(POS. 5.6) *Current output value in case of failure* **[mA V.alarm =% XXX]** **AL3 [OCACV]**

The output current signal can be specified by the user in case of failure of either, empty pipe, coils interrupted, or ADC error. The signal current is set as a percentage (0 to 125%) of the 0/4-20mA current. 125% corresponds to 24mA and does not depend on the selected range (0-20/4-20mA).

The NAMUR NE43 recommendation asks for a alarms signalling value for the current output lower than 3.6mA (<18%) or bigger than 21mA (>105%). It would then be preferable to set the value of this function at the 10%, so that the current value in case of the a.m. cases would be 2 mA, allowing the following diagnostics:

- ☐ current < 2mA - 5%: line interrupted, power supply failure or faulty converter;
- ☐ 2mA -5% * current * 2mA + 5%: hardware alarm;
- ☐ 4mA * current * 20mA: normal working range;
- 20mA < current * 22mA: out of range, measure above 100% f.s.

(POS. 5.7) *Frequency output value in case of failure* **[Hz V.alarm=%XXX]** **AL3 [OFACV]**

This function is active with POS. 7.1 see page 29 and POS. 7.2 see page 29 enable to (FREQ.+, FREQ.-, FREQ.±) To set the frequency value assigned to the on/off output in one or more of the following failure cases:

- ☐ Empty pipe; Coils interrupted ; ADC error. Allowable range is from 0 to 125% of the frequency full scale value. Although there are no specific rules regulating cases such as these, it would be convenient to use the failure information as follows:
- ☐ 0% Hz * frequency * 100% f.s.: normal working range;
- ☐ 100% f.s. < frequency * 210% f.s.: overflow, measure above the 100% of the f.s.;
- ☐ 115% f.s. * frequency * 125% f.s.: hardware alarm condition.

MENU 6 - INPUTS

WARNING!**THIS MENU IS NOT ENABLED WHEN THE BATCHING FUNCTION IS ACTIVE.****(POS. 6.1-2) *T.direct(pos)flowtot./part reset enable* [T/P+/RESET=ON/OFF] AL3 [VTTPE] [VTPPE]**

When one of this function is enabled, the related totalizer + may be reset through the on/off input.

(POS. 6.3-4) *T.rev.(neg)flowtot./part reset enable* [T/P-/RESET=ON/OFF] AL3 [VTTNE] [VTPNE]

When one of this function is enabled, the related totalizer - may be reset through the on/off input.

(POS. 6.5) *Totalizers counting lock command* [COUNT LOCK= ON/OFF] AL3 [TCLIE]

Totalizers counting lock command enable. When this function is active, applying a voltage on the on/off input terminals the system stops the totalizers no matter which is the flow rate.

(POS. 6.6) *Measure zero lock input enable* [Meas.lock=ON/OFF] AL3 [MSLIE]

When this function is active (ON), applying a voltage on the on input terminals, the measurement is stopped, the meter will display zero flow.

(POS. 6.7) *Calibration Func. input enable* [Calibration=ON/OFF] AL3 [CALIE]

When this function is active, applying a voltage on the on/off input terminals the meter performs a autozero calibration cycle. ATTENTION: If the voltage pulse is less than 1 sec., the meter performs a calibration cycle to compensate possible thermal drifts. If the voltage pulse is more 1 sec, the meter performs a zero calibration measure. To perform the calibration it is absolutely necessary for the sensor to be full of liquid and that the liquid is perfectly still. Even very small movement of the liquid may affect the result of the calibration, and, consequently, the accuracy of the system.

(POS. 6.8) *Flow rate range change inp.* [RANGECHANGE=ON/OFF] AL3 [SRCIE]

Range change external command enable. When this function is enabled, applying a voltage on the on/off input terminals the meter switches to the second measuring range (Fs2).

N.B.: the autorange doesn't allow to use the manual range change see (POS. 4.4 see page 28).

MENU 7 - OUTPUTS

(POS. 7.1) Output 1 function selection**[Out1=XXXXXX]****AL3 [OUT1F]**

Function choice corresponding to digital Output 1. The functions are listed in the table below.

(POS. 7.2) Output 2 function selection**[Out2=XXXXXX]****AL3 [OUT2F]**

Function choice corresponding to digital Output 2. The functions are listed in the table below.

FUNCTIONS FOR OUTPUTS 1 AND 2

- ☐ **OFF:** DISABLE
- ☐ **MAX AL. +:** MAX DIRECT FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MIN AL. +:** MIN DIRECT FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MAX/MIN+/-:** MAX/MIN DIRECT FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MAX AL.-:** MAX INVERSE FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MIN AL.-:** MIN INVERSE FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **MAX/MIN-:** MAX/MIN INVERSE FLOW RATE OUTPUT (ENERGIZED = AL. OFF)
- ☐ **P.EMPTY:** EMPTY PIPE ALLARM OUTPUT (ENERGIZED = AL. OFF)
- ☐ **HARDWARE AL.:** SUM OF ALL ALARMS "energized interrupted " AND "error input signal "
- ☐ **OVERFLOW:** OUT OF RANGE ALLARM OUTPUT (ENERGIZED = FLOWRATE OK)
- ☐ **ALL ALARMS:** SUM OF ALL ALARMS POSSIBLE
- ☐ **EXT. COMM:** OUTPUT MAY TAKE A STATE DEPENDENT ON AN EXTERNAL CONTROL REMOTELY VIA MCP
- ☐ **FLOW RATE SIGN.:** FLOW DIRECTION (ENERGIZED WHEN FLOW IS NEGATIVE)
- ☐ **SCALE:** INDICATION SCALE
- ☐ **FREQ.+::** FREQUENCY POSITIVE FLOWRATE
- ☐ **FREQ.-:** FREQUENCY NEGATIVE FLOWRATE
- ☐ **FREQ.+/-:** FREQUENCY POSITIVE/NEGATIVE FLOWRATE
- ☐ **PULSES.+::** PULSE POSITIVE FLOW RATE
- ☐ **PULSES.-:** PULSE NEGATIVE FLOW RATE
- ☐ **PULSES+/-:** PULSE NEGATIVE/POSITIVE FLOW RATE
- ☐ **BATCHING:** IT'S POSSIBLE TO SET A QUANTITY OF VOLUME THAT HAVE TO BE DELIVERED DURING AN ACTUATOR (VALVE) OPENING-CLOSING CYCLE

BATCHING

Through this function, it's possible to set the volume quantity that must be delivered, managing the opening and the closing of the valve.

To activate it, follow these instructions:

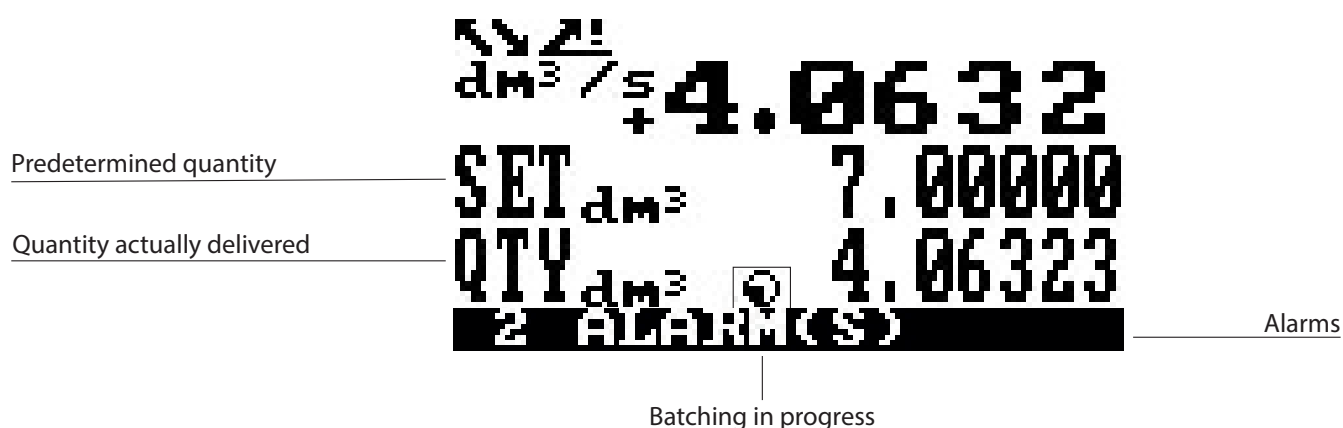
OUTPUTS

```

Out1= BATCHING
Out2= OFF
Out mA1=4_22 +/-
Out mA2=4_22 -0+
A1S=dm³/s 5.0000
A2S=dm³/s 5.0000
t 1 function selectio
    
```

1. Enter to the main menu;
2. Select Output menu;
3. Set the Out1 function on "BATCHING"

With batching activated, the display visualizes one main page:



The batching parameters settings can be customized according to what the instrument use and application is thought to.

FUNCTION	FUNCTION N° POS.	SETTING
Cut-off	"4.2" pag28	It must be set so that the flow is stably zero when the valve is closed
Damping	"4.1" pag28	It must be set on the minimum value allowed. If T batching ≤ 1s, filters must be disabled

BATCHING QUANTITY SETTINGS:

By display
By MCP command
By ModBus interface

Enter to the 3. Scale menu;
Set the B.qty (batching quantity) value

SCALES
FS1=dm³/s 5.0000
B.qty=1.00000
Batching quantity

"BTQTY?"
Through this command it can be set both the unit measure or the quantity (see MCP manual)

VALUES:
0:'ml'|1:'cm3'|2:'l'|3:'dm3'|4:'dal'|5:'hl'
|6:'m3'|7:'Ml',0.00001 <> 99999.9

Using the ModBus command 16 (see MODbus manual)

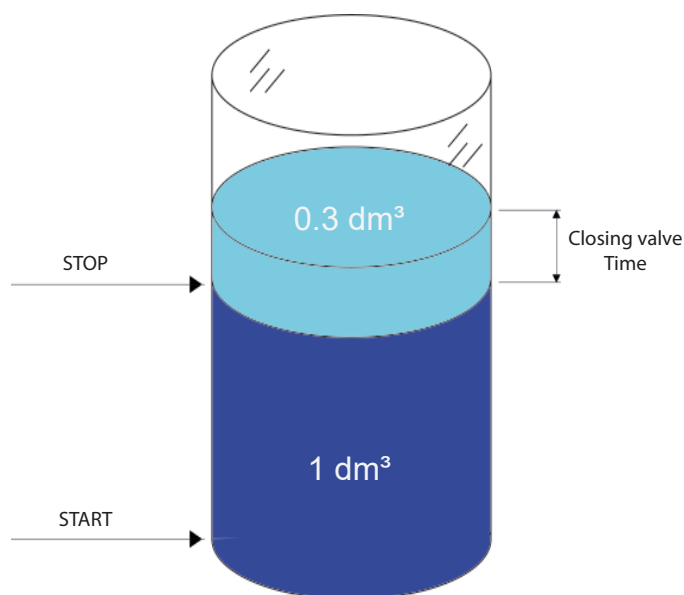
BATCHING OPERATIONS:

ACTION:	OPERATION:	NOTES:
Start/stop by display	Press "UP" key on the keyboard	When the batching process is manually stopped, the OUT1 function is disabled and the batching cycle is interrupted with the valve closing. Every time the batching cycle starts, the set quantity counter and the alarms are reset.
Start/stop by MCP	Type MCP command "BTACT"	
Start/stop by ingresso	External signal connected to digital input 1. N.B: The signal must be a pulse and its duration must be greater than the time of the measurement cycle. See "DIGITAL INPUT ON/OFF OPERATION" page 15	
Start/stop by ModBus	Send the MODbus command 5 (see MODbus manual)	
Visualize batching status by display	Batching in progress symbol on the display 	
Visualize batching status by MCP	Type the MCP command "BTACT?" If = 0 the actuator is closed If = 1 the actuator is open (batching in progress)	

The batching process automatically stops the valve when the batching quantity reaches the set value. If the delivered volume doesn't match to the required one, It is necessary to notice empirically the value to set. In this way It is possible to compensate any "delays" in the process after the stop signal.

EXAMPLE WITH DELIVERED VOLUME = 1 dm³

COMPANSATED BATCHING



NOT COMPASATED BATCHING

